Sampling methods — quick guide

1.3.1.2 Market research · Unit 1 · Mr. Akram — Akademy

Worked context: a school canteen decides whether to sell a healthier lunch. Population = 1,000 students (600 Year 12, 400 Year 13). Sample needed = 100.

Method	How the 100 are selected	Supports the decision	Limitation
Random	Number all 1,000 and use a random-number generator to pick 100; everyone has an equal chance.	Reduces researcher bias; a fair estimate of overall demand.	May include too many Year 12 by chance; needs a complete list.
Quota	Target 60 Year 12 and 40 Year 13; approach available students until each target is filled.	Quick; includes both year groups so demand can be compared.	Researcher picks the easiest to approach, so findings may be biased.
Stratified	Split register into Year 12/13; randomly select 60 and 40 to match population proportions.	Proportionate AND random — most accurate stock/marketing decision.	Takes longer; needs accurate population information.

Remember

- **Random** = everyone has an equal chance.
- **Quota** = correct group numbers, but available people are chosen (convenience).
- **Stratified** = correct group proportions AND random selection within each group.

Judgement

Stratified sampling is usually the most representative. Quota may be better when a quick, low-cost decision is needed.